Engineering - Undergraduate Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Snubber circuit is used with SCR
   (a) either series or parallel.
   (b) in series.
   (c) in an alternating manner.
   (d) in series-parallel.
   (e) in anti series.

2. Find the inverse Laplace transform f(t) = L^-1 {log [(s + 1) / (s - 1)]}, s > 1.
   (a) (1 - e^(-2t)) / t
   (b) (2 sinh t) / t^2
   (c) (2 tanh t) / t
   (d) (2 sinh t) / t
   (e) (2 tanh t) * t

3. Find f_1(t) {_*} f_2(t) if f_1(t) = 2e^-4tu(t) and f_2(t) = 5 cos 3t u(t).
   (a) f_1(t) {_st} f_2(t) = (1.6 cos 3t + 1.2 sin 3t - 1.6e^-4t)u(t)
   (b) f_1(t) {_st} f_2(t) = (1.2 cos 3t + 1.6 sin 3t - 1.3e^-4t)u(t)
   (c) f_1(t) {_st} f_2(t) = (1.4 cos 3t + 1.2 sin 3t - 1.5e^-4t)u(t)
   (d) f_1(t) {_st} f_2(t) = (1.6 cos 3t + 1.0 sin 3t - 1.6e^-4t)u(t)

4. Evaluatedivuat the point (- 1, 1, 2) foru= x^2i+e^xyj+ xyzk(1)
   (a) - 3 - e^1
   (b) - 2 - e^-1
   (c) 2 + e^-1
   (d) - 3 - e^-1
   (e) -3 + e^1

5. A Lissajous pattern on an oscilloscope has 5 horinzontal tangencies and 2 vertical tangencies. Th
   will be
   (a) 750 Hz.
   (b) 250 Hz.
   (c) 625 Hz.
   (d) 500 Hz.
   (e) 800 Hz.

True / False (10 marks)

Answer True or False

6. True or False: The inverse Laplace transform of F(s) = [(2s^2 - 4) / {(s - 2)(s + 1)(s - 3)}] is f(t) =
7. True or False: The correct answer to 'Evaluate the determinant |1- 4- 5| Δ=|123| |- 31- 2|' is '-28'.
8. True or False: The absolute pressure of a system with a gauge pressure of 1.5 MPa and an atmo
9. True or False: The mean-square quantization error for a uniformly distributed signal quantized wi
10. True or False: The theoretical variance of the given probability distribution is 0.625.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. An induction motor has a rotor current of 10 amperes per phase with a slip of 5% and an effective rotor resistance of 0.1 ohm per phase. Calculate the internal power developed per phase and discuss the implications of rotor resistance on motor performance.
12. Given an input signal v(t) = I(t) passed through a filter with the transfer function (1 - e^(-j))/j, analy your findings.

End of Paper